7.1 Constant field scaling allows us represent changes in the scale of CMOS devices with a unitless quantity S. Assuming ideal constant field scaling (that is, scaling the size of a CMOS device without changing the strength of its internal electric field), we may use Table 7.4 to determine that gate delay τ scales by a factor of $\frac{1}{S}$. Thus, using an initial gate delay of $\tau = 500$ ps, we may assume that by scaling the device in question by a factor of $S = 2$, the new gate delay $\tau = \frac{500}{2} = 250$ ps.

Using Table 7.5, we see that repeated wire RC delay per unit length, assuming constant field scaling of gates, scales by a factor of $\sqrt{S}$; in our case, this causes the delay to *increase*, so that $t_w r = 500\sqrt{2} \approx 707$ ps. This concords with our intuitive sense of wire scaling and its impact on delay; reduced height in an interconnect will lower its cross-sectional area and increase its resistance, leading to higher delay.

In total, this leads to an overall reduction in delay from $500 + 500 = 1000$ ps to $250 + 707 = 957$ ps, or a 4.3% reduction in delay.

7.2 First, the worst-case input delay for the NOR2X1 gate, based on the TSMC databook, $B \rightarrow Y \uparrow$ is 45 ps, and best-case input delay $A \rightarrow Y \downarrow$ is 18 ps. We will use the formula given in the TSMC databook for adjusting delay based on derating factors

$$t_{\text{TPD}} = (K_{\text{process}} \cdot [1 : e + 35 + (K_{\text{volt}}) \cdot \Delta V_{dd})] \cdot [1 + (K_{\text{temp}} \cdot \Delta T)] \cdot t_{\text{typical}}$$

and using the allowable voltage and temperature ranges found in the databook, $1.62 - 1.98$ V and $0 - 125^\circ$ C, respectively, we may determine worst-case and best-case delay as follows:

(a) At $V_{DD} = 1.62$ V and $T = 125^\circ$ C, with a slow process speed,

$$t_{\text{TPD}} = 1.293 \cdot [1 + (-0.731 \cdot -0.18)] \cdot [1 + (0.00123 \cdot 100)] \cdot t_{\text{typical}} = 73.9 \text{ ps.}$$

(b) At $V_{DD} = 1.98$ and $T = 0^\circ$ C, with a fast process speed,

$$t_{TPD} = 0.781 \cdot [1 + (-0.511 \cdot 0.18)] \cdot [1 + (0.00123 \cdot 0)] \cdot 0.018 = 12.8 \text{ ps.}$$

7.2 Using Eq. 7.3,

$$C = \frac{I_{\text{dc}-\text{max}}}{\alpha V_{DD} f},$$